MAT499 / 599

Project 1: Exploring the learning of Fully-connected Neural Networks.

February 6, 2026

DUE: 11:59pm on Monday February 16th, 2026

Submission: Please submit this project to canvas as an iPython Notebook (.ipynb file). The project is worth 25% and 20% of the overall grade for MAT499 and MAT599, respectively. While the submission is a python script, also focus on presenting the questions logically (write them out!), explaining what the code in each block does using comments, and answering the word questions fully in text blocks within the notebook.

Task: This project will attempt to model the red wine quality dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/186/wine+quality). A csv file of this data is available on the GitHub for this course (https://raw.githubusercontent.com/benjaminmlucas/MAT499/refs/heads/main/module_1/winequality.csv). This dataset provides various characteristics of red wines and attempts to predict the quality score of the wine based on these characteristics.

Prior to running the following set a seed to 499 or 599 (your enrolled class) so that your results are the same for every run of your notebook. You will need to set the seed for the numpy.random, random, and torch libraries.

Question 1

Fit the following 3 different models to the data to predict wine quality:

1. Linear Model
2. A Single-hidden Layer Neural Network with 12 neurons.
3. A Three-hidden Layer Neural Network with 12, 8, and 4 neurons in the hidden layers.

Note in a text box the number of parameters to learn for each of the three models above.

Use gradient descent with a learning rate of 0.01 to learn the parameters of each model. Choose a number of epochs that is large enough to ensure that the loss has converged (let's say constant to at least the 4^{th} decimal place).

To analyze the learning of these models, we would like to produce a plot of Loss against epoch for each model. Display the plot with all three lines on the same set of axes.

In a text box, discuss: (a) whether they all converged to the same minimum loss value or not, and can you hypothesize a relationship between loss and the number of parameters in a neural network, and (b) whether the number of epochs required to converge to its minimum differed between models. Is there an apparent relationship between this and number of parameters in a neural network?

Question 2

Revisit the 3-hidden layer model you defined above. Change the number of neurons to 16, 8, and 4, for the hidden layers in that order (you may wish to define a new class, depending on how you coded the above). Learn the model with the following 5 different learning rates: 0.00001, 0.0001, 0.001, 0.01, 0.1. Create a plot of Loss against Epoch and plot each of the 5 variants on the same axis.

In a text box, discuss the impact learning rate has on: (a) the minimum loss and (b) the number of epochs required for the model to converge.

Question 3

Split the data in half at random and relearn the three models from Question 1 on this new smaller dataset. Does the quantity of data change the value of the loss that the models converge on? Does it affect the models similarly or differently? Attempt to draw a hypothesis from these results (in the case that your prior results are incorrect, your hypothesis will be graded based on your results, not its real-life validity).